IMPG1-related Recessive Retinopathy — Comprehensive Disease Characterization Report

Disease: IMPG1-related Recessive Retinopathy **Causal gene: IMPG1** (interphotoreceptor matrix proteoglycan 1, encoding SPACR/IPM150) **MONDO ID:** MONDO:1040037 · **Phenotype OMIM:** 616151 (Vitelliform macular dystrophy 4, VMD4)

Summary

IMPG1-related recessive retinopathy is a rare autosomal-recessive inherited retinal dystrophy caused by biallelic (loss-of-function or hypomorphic) variants in *IMPG1*, the gene on chromosome 6q14.1 encoding SPACR (also called IPM150), a chondroitin-sulfate proteoglycan of the interphotoreceptor matrix (IPM). This report concerns the IMPG1 gene specifically — the causal gene named in the task header — and not the clinically similar IMPG2 (IPM200) disorder or the dominant *BEST1*/*PRPH2* vitelliform dystrophies, which are important differential diagnoses. The disease was first defined by Manes and colleagues, who identified both dominant and recessive *IMPG1* families within a vitelliform macular dystrophy (VMD) cohort [PMID: 23993198].

Mechanistically, SPACR is one of the two principal proteoglycans (with IMPG2/IPM200) that build and organize the IPM — the specialized extracellular matrix that ensheaths rod and cone photoreceptor outer segments and provides the chondroitin-sulfate–rich cone glycocalyx. Loss or dysmaturation of SPACR, or its mislocalization into the subretinal space, disrupts the IPM glycocalyx, interferes with photoreceptor outer-segment/connecting-cilium development, and leads to accumulation of subretinal "vitelliform" material. Clinically this manifests as adult-onset, slowly progressive, bilateral vitelliform maculopathy with drusen-like deposits and cone-predominant photoreceptor loss. The electro-oculogram (EOG) is characteristically normal or only mildly reduced, distinguishing it from Best disease.

The disorder is ultra-rare, accounting for a small minority of VMD families. Onset is typically in the fourth decade with moderate visual impairment (mean visual acuity ~20/40). There is no sex predilection and the disease does not affect survival. No approved disease-specific or curative therapy exists; management is supportive (low-vision rehabilitation) together with genetic counseling, carrier/cascade testing, and reproductive options. Because the recessive disease is driven by biallelic loss of function, gene-augmentation is a rational future therapeutic strategy, though no IMPG1 clinical trials currently exist.

Key Findings

Finding 1 — IMPG1 (encoding SPACR) is the causal gene; biallelic variants cause recessive vitelliform retinopathy

The foundational genetic evidence comes from a large VMD cohort in which *IMPG1* (6q14, encoding the interphotoreceptor-matrix proteoglycan SPACR/IPM150) was shown to cause disease in both dominant and recessive forms. For the **autosomal-recessive** form, two affected individuals from one family were homozygous for the splice-site variant **c.807+1G>T**, and two from a second family were compound heterozygous for **c.461T>C (p.Leu154Pro)** and **c.1519C>T (p.Arg507*)** [PMID: 23993198]. The verbatim report states: *"Two individuals from one family affected by autosomal-recessive VMD were homozygous for the splice-site mutation c.807+1G>T, and two from another family were compound heterozygous for the mutations c.461T>C (p.Leu154Pro) and c.1519C>T (p.Arg507(∗))."* The same work establishes the gene product: *"IMPG1 encodes the SPACR protein, a component of the rod and cone photoreceptor extracellular matrix domains."*

This provides the identity link between the clinical disease (MONDO:1040037) and its molecular cause. The recessive variant spectrum includes splice-site, missense, and nonsense alleles — consistent with a loss-of-function/hypomorphic mechanism.

Finding 2 — Mechanism: SPACR is an IPM proteoglycan whose loss/dysmaturation disrupts the matrix and causes subretinal vitelliform deposits

SPACR/IMPG1 and IMPG2 are the two major proteoglycans of the interphotoreceptor matrix, the principal source of chondroitin sulfate and constituents of the cone glycocalyx (peanut-agglutinin/PNA-positive) [PMID: 32265257]. The review by Salido & Ramamurthy states directly: *"The IPM includes the following two unique proteoglycans: IPM proteoglycan 1 (IMPG1) and IMPG2. Patients with mutations in IMPG1/IMPG2 develop visual deficits with subretinal material accumulation, highlighting the critical role of the IPM in vision."*

Both proteins carry **SEA domains** whose autoproteolytic cleavage is required for maturation; disease-associated human mutations in the SEA domain inhibit this proteolysis [PMID: 36109576]: *"Point mutations mimicking human mutations in the SEA domain of IMPG1 that are associated with vision disease inhibited proteolysis."* At the biochemical level, SPACR/Spacrcan binds hyaluronan and chondroitin sulfate via RHAMM-type hyaluronan-binding motifs, which is how it organizes the IPM [PMID: 15044457]: *"These RHAMM-type HABMs in Spacrcan are involved in hyaluronan binding."*

Together these establish a coherent lesion: defective SPACR (through absent protein, impaired SEA-domain maturation, or lost glycosaminoglycan binding) fails to build/organize the IPM glycocalyx around outer segments, and subretinal material accumulates as the clinically observed vitelliform deposit.

Finding 3 — Clinical phenotype: adult-onset, moderate, slowly progressive vitelliform maculopathy with drusen-like deposits

In the Meunier cohort, patients with IMPG1 (or IMPG2) mutations had **late onset and moderate visual impairment**: *"Patients with IMPG1 or IMPG2 mutations had a late onset and moderate visual impairment (mean visual acuity, 20/40; mean age of onset, 42 years)"* [PMID: 25085631]. The presumed-recessive sporadic case had onset at **38 years and acuity 20/50**: *"even in the sporadic case of family 3 with a presumed recessive transmission (age at onset, 38 years; mean visual acuity, 20/50)."* Drusen-like lesions adjacent to the vitelliform deposits were observed, and most cases had a normal or only moderately reduced EOG Arden ratio [PMID: 25085631;

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The disease is a rare cause of VMD overall.

Finding 4 — Dual inheritance (AD and AR) and a phenotypic spectrum spanning VMD to retinitis pigmentosa

IMPG1 is one of a small set (~9) of **dual-inheritance** inherited-retinal-dystrophy (IRD) genes, in which both monoallelic (dominant) and biallelic (recessive) disease occur. In the Portuguese IRD-PT registry, dual-inheritance genes accounted for 12% of genetic diagnoses, with phenotype modulated by variant location and allele dosage [PMID: 42321002]: *"This spectrum, modulated by variant location and allele dosage, determines phenotypes and contributes to IRD heterogeneity."* Beyond classic VMD, monoallelic or biallelic *IMPG1* variants have been associated with **retinitis pigmentosa (RP)** [PMID: 32817297]: *"This study discusses a previously unreported association between monoallelic or biallelic"* IMPG1 variants and RP. The original gene description already included recessive VMD families [PMID: 23993198]. The practical implication: the IMPG1 recessive phenotype is best viewed as a spectrum from focal macular vitelliform disease to more widespread outer-retinal/RP-like degeneration.

Finding 5 — Model-organism evidence: pathology requires IMPG1 mislocalization, not simple absence

A key mechanistic insight comes from mouse genetics. **Impg1-knockout mice have a normal retina**: *"No evident morphologic or functional deficits were found in mice lacking IMPG1"* [PMID: 32265257]. However, in **IMPG2-knockout** mice, IMPG1 abnormally accumulates in the subretinal space, forming subretinal lesions and reducing visual function: *"In the absence of IMPG2, IMPG1 abnormally accumulated at the subretinal space."* Critically, **double IMPG1/IMPG2-knockout** mice have normal structure and function — demonstrating that **aberrant IMPG1 distribution, rather than its mere loss, is the main driver of the visual alteration** in that model. Complementary metabolomics in IMPG2-deficient retinas with subretinal lesions showed *"reduced glutamate metabolism, urea cycle, and galactose metabolism suggesting affected energy metabolism"* [PMID: 37440072]. This nuance — that mislocalized matrix protein is pathogenic — is important for interpreting human disease and for therapy design.

Finding 6 — Diagnosis relies on multimodal retinal imaging

IMPG1 vitelliform lesions are **subretinal on OCT** and **hyperautofluorescent** on short-wavelength fundus autofluorescence, with genetic panel testing confirming the diagnosis [PMID: 40315441]: *"Short-wavelength fundus autofluorescence confirmed that the lesions were hyperautofluorescent."* Multimodal **adaptive-optics** imaging of VMD showed within-lesion cone density reduced to 37% of normal and RPE to 8.4% of normal, and — importantly for differential diagnosis — *"cones were affected to a greater extent than RPE in patients with*

IMPG1 and IMPG2 pathogenic variants, but RPE was affected more than cones in BEST1 and PRPH2 VMD" [PMID: 35900727]. A heterozygous p.Leu154Pro carrier showed outer-retinal ellipsoid/interdigitation-zone disruption on OCT [PMID: 30688845]: *"Optical coherence tomography revealed disorganization and decreased reflectance of the foveal ellipsoid and interdigitation zones."* The EOG Arden ratio is typically normal or mildly reduced, which helps separate IMPG1 disease from Best disease (BEST1) [PMID: 23993198].

Finding 7 — IMPG1 is required for photoreceptor outer-segment/connecting-cilium development and IPM formation; variable expressivity with modifiers

A developmental role was demonstrated in human pluripotent stem-cell-derived retinal organoids: blocking IMPG1 (and CD44) *"affected the development of photoreceptors, their inner/outer segments and connecting cilia and disrupted IPM formation, with IMPG1 having an earlier and more significant impact"* [PMID: 29777959]. Clinically, siblings carrying an identical *IMPG1* exon-7 deletion showed divergent phenotypes (one with a single foveal lesion, the other with asymptomatic multifocal extrafoveal lesions), supporting **variable expressivity with individual modifiers** [PMID: 40315441]: *"Intrafamilial variability in the topography of the lesions with multifocality in one of the siblings suggests retina-wide abnormalities with individual modifiers modulating disease expression."*

Finding 8 — Consolidated identifiers, classification, and management

Attribute	Value
Gene symbol	**IMPG1**
HGNC	HGNC:6055
NCBI Gene	3617
Ensembl	ENSG00000112706
UniProt	Q17R60 (SPACR / IPM150)
Cytoband	6q14.1
Gene OMIM	602870
Disease MONDO	MONDO:1040037
Phenotype OMIM	VMD4, 616151
ICD-10	H35.5 (hereditary retinal dystrophy)
Mouse ortholog	*Impg1* (NCBI Gene 63859; MGI:1926876)
Inheritance	Autosomal recessive (biallelic LoF/hypomorphic); IMPG1 is a dual-inheritance gene

The disease is ultra-rare: *"IMPG1 was the causal gene in 3 families (IMPG1 1-3, 11 patients) and IMPG2 in a fourth family (2 patients)"* in the Meunier cohort [PMID: 25085631]. There is no sex predilection; onset is in adulthood; the course is slowly progressive; and there is no effect on survival. Management is supportive (low-vision rehabilitation) plus genetic counseling, carrier/cascade testing, and prenatal/preimplantation genetic testing options. Dual inheritance is confirmed [PMID: 23993198]: *"We conclude that IMPG1 mutations cause both autosomal-dominant and -recessive forms of VMD."*

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *IMPG1* variant** (splice-site c.807+1G>T; nonsense p.Arg507*; missense p.Leu154Pro; or exon deletion) → **leads to** absent, truncated, or dysfunctional SPACR/IPM150 protein (loss-of-function / hypomorphic). *[Demonstrated genetically —*

   P *23993198*

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2. Loss of functional SPACR → **results in** failure of SEA-domain proteolytic maturation and loss of hyaluronan/chondroitin-sulfate binding, so the protein cannot assemble into the matrix. *[Demonstrated in vitro —*

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   P *15044457*

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3. Defective SPACR assembly → **disrupts** the chondroitin-sulfate–rich interphotoreceptor matrix glycocalyx that ensheaths rod and cone outer segments. *[Demonstrated —*

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4. **Branch A (developmental):** IPM disorganization during retinogenesis → **impairs** photoreceptor inner/outer-segment and connecting-cilium formation. *[Demonstrated in retinal organoids —*

   P *29777959*

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5. **Branch B (homeostatic/adult):** IPM disorganization and mislocalized matrix proteoglycan → **cause** accumulation of subretinal "vitelliform" material and separation/stress at the photoreceptor–RPE interface. *[Inferred from human imaging + supported by mouse mislocalization model —*

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   P *40315441*

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6. Subretinal deposit + outer-segment compromise → **produces** cone-predominant photoreceptor dysfunction and loss (cones affected more than RPE). *[Demonstrated by*

adaptive-optics imaging —

P *35900727*

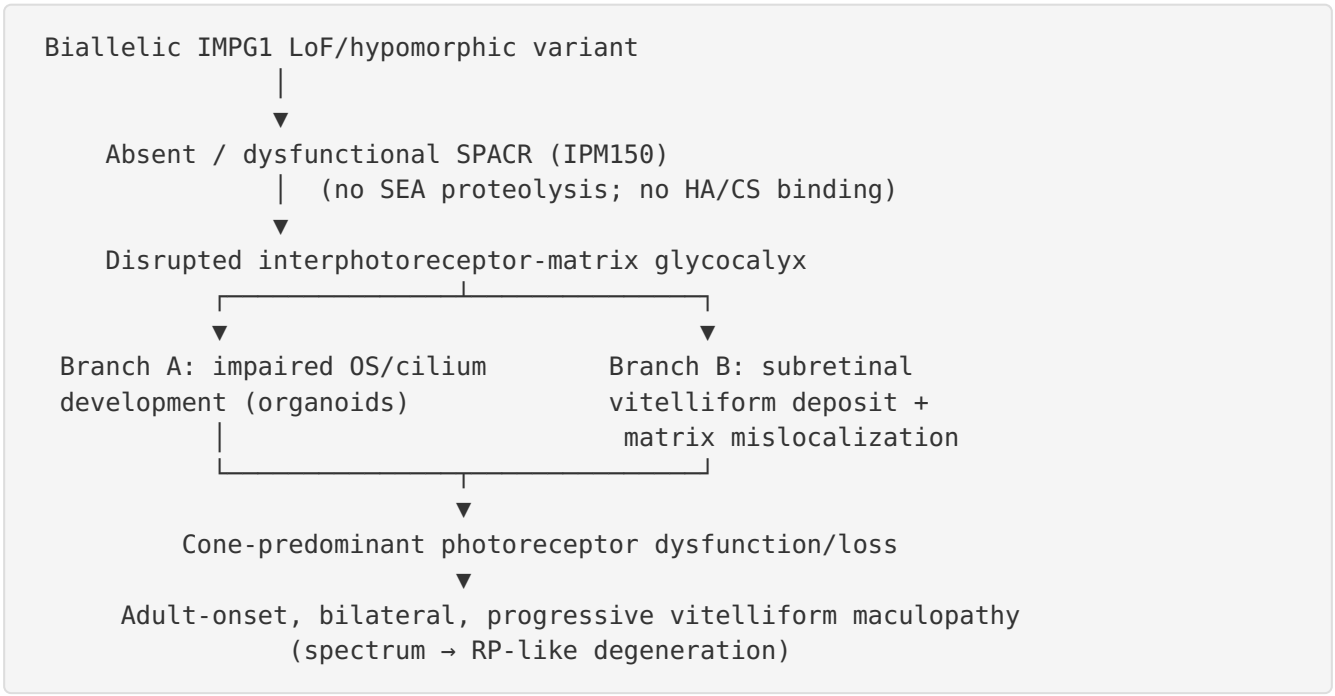
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7. Progressive photoreceptor loss → **manifests** as adult-onset, bilateral, slowly progressive vitelliform maculopathy with moderate visual-acuity loss, metamorphopsia, and (in the broader spectrum) RP-like changes. *[Demonstrated clinically —*

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Upstream vs downstream: The *IMPG1* mutation and SPACR dysfunction are the upstream lesion; IPM disorganization is the proximate intermediate; subretinal deposition and cone loss are downstream effectors of visual loss. **Cell types involved:** rod and cone photoreceptors (CL:0000573 retinal cone cell; CL:0000604 retinal rod cell), with the retinal pigment epithelium (CL:0002586) secondarily involved. **Anatomy (UBERON):** retina (UBERON:0000966), specifically the macula/fovea (UBERON:0005396 macula lutea) and the interphotoreceptor/subretinal space; photoreceptor outer segment (GO:0001750) and connecting cilium are the key subcellular sites. **Biological processes (GO):** extracellular matrix organization (GO:0030198), photoreceptor cell development/morphogenesis, visual perception (GO:0007601), glycosaminoglycan binding (GO:0005539). **Chemical entities (CHEBI):** chondroitin sulfate (CHEBI:37397), hyaluronic acid (CHEBI:16336).

A central interpretive point from the mouse work is that in the *Impg2*-null model, **mislocalized IMPG1** is pathogenic, whereas simple *Impg1* loss is tolerated in mouse. In humans, however, biallelic *IMPG1* variants clearly cause disease — implying either species differences in IPM redundancy or that human pathogenic alleles produce a mislocalized/dysfunctional product rather than a null state. This distinction matters for therapy: a pure gene-replacement approach assumes a loss-of-function target, which fits the human recessive LoF alleles but should be reconciled with the mislocalization mechanism.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
23993198	*Mutations in IMPG1 cause vitelliform macular dystrophies*	Human genetic	Defines causal gene; specific recessive variants; dual inheritance (F1, F4, F8)
25085631	*Frequency and clinical pattern of VMD caused by IMPG1/IMPG2*	Human clinical cohort	Onset, acuity, rarity, EOG (F3, F8)
32265257	*Proteoglycan IMPG2 shapes the IPM and modulates vision*	Review + mouse	IPM biology; Impg1-KO normal; mislocalization mechanism (F2, F5)
36109576	*IMPG1/IMPG2 proteolyze in the SEA domain…*	In vitro	SEA-domain proteolysis; disease mutations inhibit it (F2)
15044457	*Spacrcan binding to hyaluronan and other GAGs*	Biochemical	HA/CS binding function of SPACR (F2)
29777959	*ECM expression in hPSC retinal organoids…*	In vitro (organoid)	Developmental role in OS/cilium/ IPM formation (F7)
32817297	*Pathogenic variants in IMPG1…*	Human genetic	IMPG1 variants associated with RP (F4)
42321002	*IRD genes with dual inheritance: IRD-PT registry*	Human registry	Dual-inheritance concept; allele dosage modulation (F4)
37440072	*Retinal metabolic profile in IMPG2-deficiency mice*	Mouse metabolomics	Energy-metabolism changes with subretinal lesions (F5)
40315441	*Intrafamilial variability of IMPG1 vitelliform dystrophy*	Human clinical	Imaging; variable expressivity/ modifiers (F6, F7)
35900727	*Photoreceptor–RPE relationships in VMD by adaptive optics*	Human imaging	Cone-predominant loss distinguishing IMPG1/2 (F6)
30688845	*Early-onset vitelliform dystrophy & p.Leu154Pro*	Human case	OCT outer-retinal disruption (F6)

Supporting/context papers reviewed: IMPG2 genomic organization and candidacy [PMID: 11726612]; gene-discovery prevalence survey confirming IMPG1 as a rare VMD gene [PMID: 24702842]; RP25-locus candidate-gene analysis [PMID: 16354621]; additional AVMD case reports [PMID: 37661650].

Consistency of the evidence: Human genetic (23993198, 32817297), clinical/imaging (25085631, 40315441, 35900727, 30688845), in-vitro biochemistry (36109576, 15044457), organoid developmental biology (29777959), and mouse genetics/metabolomics (32265257, 37440072) converge on a single IPM-centered mechanism. The main internal tension is the discordant mouse Impg1-KO phenotype (normal) versus clear human disease, addressed in the Interpretation section.

Section-by-Section Synthesis (Knowledge-Base Fields)

- **1. Disease information:** Rare autosomal-recessive macular/retinal dystrophy; MONDO:1040037; OMIM VMD4 616151; ICD-10 H35.5. Synonyms: IMPG1-related vitelliform macular dystrophy, autosomal-recessive VMD4, SPACR/IPM150 retinopathy. Evidence is from aggregated disease-level resources and published patient cohorts/case series, not EHR.
- **2. Etiology:** Purely genetic — biallelic *IMPG1* variants (splice, nonsense, missense, exon deletion). No environmental/infectious cause. Consanguinity facilitates recessive homozygous cases; IMPG1 is a dual-inheritance gene where allele dosage and variant location modulate phenotype. No established protective alleles or gene–environment interactions.
- **3. Phenotypes:** Decreased central visual acuity (HP:0000572 visual loss; HP:0007663 progressive macular degeneration), metamorphopsia, vitelliform macular lesion/subretinal deposits (HP:0011505 macular dystrophy), drusen-like deposits, outer-retinal (ellipsoid/interdigitation-zone) disruption, and in the broader spectrum pigmentary retinopathy (HP:0000510). Onset adult (4th decade), severity moderate (mean 20/40), course slowly progressive, bilateral. Quality-of-life impact is moderate — central/reading vision affected, peripheral vision/mobility relatively spared in the macular form.
- **4. Genetic/molecular:** Causal gene IMPG1 (HGNC:6055, Gene 3617). Recessive alleles: c.807+1G>T (splice), c.461T>C p.Leu154Pro (missense), c.1519C>T p.Arg507* (nonsense), exon-7 deletion. Functional consequence: loss of function / hypomorphic with SEA-domain maturation defect. Modifiers inferred from intrafamilial variability. No epigenetic or chromosomal mechanisms described.

- **5. Environmental:** None established — monogenic disorder.
- **6. Mechanism:** See causal chain above (IPM proteoglycan loss → glycocalyx disruption → subretinal deposit + cone-predominant loss).
- **7. Anatomy:** Retina (UBERON:0000966), macula/fovea (UBERON:0005396); photoreceptor outer segment (GO:0001750) and interphotoreceptor/subretinal space; rod and cone photoreceptors (CL:0000604, CL:0000573); RPE secondarily (CL:0002586). Bilateral.
- **8. Temporal development:** Adult onset (~38–42 y), insidious/chronic, slowly progressive, lifelong; no remission; potential early window for intervention before photoreceptor loss.
- **9. Inheritance/population:** Autosomal recessive; ultra-rare (3/49 VMD families IMPG1-related); no sex predilection; carrier frequency not precisely established (ultra-rare). Penetrance appears high but expressivity variable.
- **10. Diagnostics:** Multimodal imaging (OCT subretinal vitelliform lesion, hyperautofluorescent on SW-FAF, cone-predominant loss on adaptive optics), normal/mildly reduced EOG, and molecular confirmation by IRD gene panel / WES / WGS / single-gene testing. Differential: BEST1 (abnormal EOG, RPE-predominant), PRPH2, and age-related macular degeneration.
- **11. Prognosis:** Chronic, slowly progressive central vision loss; no effect on survival; legal blindness uncommon but reading/driving vision commonly impaired.
- **12. Treatment:** No approved disease-specific therapy. Supportive low-vision rehabilitation (NCIT: Low Vision Aid / rehabilitation therapy). Gene-augmentation is a rational future strategy given biallelic LoF, but no IMPG1 trials exist.
- **13. Prevention:** No primary prevention. Genetic counseling, carrier/cascade testing, prenatal diagnosis, and preimplantation genetic testing are the relevant interventions.
- **14. Other species:** Mouse ortholog *Impg1* (Gene 63859; MGI:1926876); no major naturally occurring animal disease documented in the reviewed literature.
- **15. Model organisms:** *Impg1*-KO mouse (normal retina), *Impg2*-KO mouse (subretinal IMPG1 mislocalization/lesions), double-KO (normal); human iPSC-derived retinal organoids for developmental IPM studies.

Limitations and Knowledge Gaps

- **Small evidence base.** The disease is ultra-rare; recessive genotype–phenotype correlations rest on a handful of families and individual case reports, limiting precision on penetrance, expressivity, and natural history.
- **Mouse–human discordance.** *Impg1*-KO mice are phenotypically normal, so there is no faithful recessive mouse model; the pathogenic mechanism in mouse is mislocalization (via *Impg2* loss), which may not fully recapitulate human IMPG1-null biology.
- **Overlap with IMPG2.** Much mechanistic and metabolomic detail derives from IMPG2 studies; direct IMPG1-specific functional data (beyond organoids and biochemistry) are limited.
- **Epidemiology unquantified.** No reliable prevalence/incidence or carrier-frequency estimates specific to recessive IMPG1 disease; gnomAD-based carrier calculations were not completed here.
- **Dominant vs recessive boundary.** As a dual-inheritance gene, the determinants (variant location, dosage, modifiers) that tip IMPG1 toward dominant vs recessive expression — and toward VMD vs RP — are incompletely defined.
- **No omics on human IMPG1 tissue.** Transcriptomic/proteomic/metabolomic profiling specific to human IMPG1 recessive retina is lacking.

Proposed Follow-up Experiments / Actions

1. **Build a faithful IMPG1 recessive model** — generate knock-in mice (or retinal organoids) carrying human recessive alleles (e.g., p.Arg507*, c.807+1G>T) to test whether human LoF produces subretinal deposits where the mouse Impg1-null does not, clarifying the mislocalization-vs-loss question.
2. **Genotype–phenotype registry** — aggregate IMPG1 recessive cases internationally (GeneMatcher/registry) with standardized OCT, FAF, adaptive optics, EOG/ERG and longitudinal acuity to define natural history and progression rate — essential endpoints for any future trial.
3. **Carrier-frequency/prevalence estimate** — mine gnomAD for biallelic-pathogenic burden of IMPG1 LoF alleles to estimate recessive disease prevalence and carrier frequency.

4. **SEA-domain functional assays** — systematically test recessive missense alleles (e.g., p.Leu154Pro) for SEA-domain proteolysis and HA/CS binding to classify VUS and refine ACMG scoring.
5. **Preclinical gene augmentation** — evaluate AAV-mediated IMPG1 delivery in the knock-in model, with attention to correct IPM targeting/maturation rather than mere expression.
6. **Modifier discovery** — exploit intrafamilial variability (identical-variant siblings with divergent phenotypes) via whole-genome sequencing to identify modifier loci (e.g., CD44, IMPG2) influencing expressivity.
7. **Differential-diagnosis biomarker** — validate the cone-predominant (vs RPE-predominant) adaptive-optics signature as a prospective classifier separating IMPG1/IMPG2 from BEST1/PRPH2 VMD.

Evidence source legend: human clinical/genetic (23993198, 25085631, 32817297, 42321002, 40315441, 35900727, 30688845); in vitro/biochemical (36109576, 15044457); organoid developmental (29777959); mouse model (32265257, 37440072).
